

Accuracy Is Not Always We Need: Precision-Aware Bayesian Yield Optimization

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Abstract—Integrated circuit yield optimization plays a vital role in ensuring reliable semiconductor manufacturing, directly impacting both product quality and production costs. Current approaches to yield optimization face two fundamental challenges that limit their practical effectiveness. First, yield estimation requires intensive computational resources. Second, traditional black-box optimization methods inefficiently allocate these resources across design candidates. Most existing approaches compound these issues by performing detailed yield estimations uniformly across all candidates, regardless of their potential quality. To address these limitations, we introduce a novel precision-aware yield optimization framework that intelligently adapts computational resource allocation based on each design candidate's predicted performance. Our approach moves beyond simple simulation counting by incorporating a Figure of Merit (FoM) as a continuous quality metric. By combining a Continuous AutoRegression model to characterize the relationship between true yield and precision levels with a sophisticated multi-fidelity acquisition strategy, our framework achieves optimal resource distribution. Experimental validation on four industry-standard benchmark circuits demonstrates that our method converges with fewer than 1,000 simulations, reducing simulation costs by over 10× while achieving better final designs and robustness than state-of-the-art high-fidelity approaches.

Index Terms—Yield Optimization, Yield Estimation, Multi-fidelity Bayesian Optimization, Continuous Autoregression

I. INTRODUCTION

As integrated circuit technology scales down toward the physical limits of devices, the impact of process variations on chip performance becomes increasingly severe, *e.g.*, doping fluctuation, intra-die mismatches, and threshold voltage variation. These variations can cause circuit performance to deviate from nominal design values, potentially failing to meet required specifications, which is particularly pronounced in analog and mixed-signal CMOS circuits. Therefore, designers must not only optimize nominal circuit performance but also ensure robustness against process variations, thereby giving rise to the yield optimization problem.

The complexity of yield optimization stems from two fundamental challenges. First, yield estimation requires extensive computational resources through circuit simulation. Second, the yield function's black-box nature prevents the application of efficient gradient-based optimization methods. Traditional yield optimization frameworks address these challenges by combining two key components: an estimator that evaluates

circuit yield with reduced simulation cost, and an optimizer that performs efficient black-box optimization. This creates an iterative process where the optimizer selects design parameters based on prior estimation results, followed by yield estimation of the selected designs.

Recent research has primarily focused on developing accurate yet efficient yield estimation methods. Despite significant advances, Monte-Carlo (MC) simulation remains the gold standard for reliability, though its computational demands are substantial. For example, verifying a 99.9999% yield requirement for a 6T-SRAM cell necessitates at least 10^7 simulations—a computational burden that proves impractical for real-world applications.

Two main approaches have emerged to reduce yield estimation costs: importance sampling and surrogate modeling. Importance sampling enhances estimation efficiency by concentrating samples in failure regions, as demonstrated by methods such as Minimal Norm Importance Sampling [1], Hyper-spherical Clustering and Sampling [2], Adaptive Importance Sampling [3], Optimal Manifold Importance Sampling [4], and Variational Importance Sampling [5]. Alternatively, surrogate modeling employs data-driven machine learning techniques to approximate SPICE simulation behavior, significantly reducing computational overhead. Notable approaches include Gaussian Process (GP) [6], [7], Conditional Normalizing Flow (CNF) [8], low-rank tensor approximation [9], Bayesian neural network [10], and radial basis function network [11].

Once the estimated yield is obtained, various black-box optimization algorithms can be used as yield optimizers, with Bayesian optimization (BO) being most widely used [6], [8], [10], [12]–[14]. However, a critical yet often overlooked observation is that the ultimate goal of yield optimization differs from that of accurate yield estimation—we primarily need to understand yield trends rather than exact values. This insight reveals that many existing methods are overdesigned, expending unnecessary computational resources on precise yield estimates for suboptimal or trend-irrelevant designs.

Several studies have attempted to address this inefficiency. [15] introduced an optimal computing budget allocation (OCBA) algorithm that adjusts resources based on MC variance. [12] and [13] employed two-stage MC estimators, allocating more simulations to promising designs. [16] developed a multi-fidelity GP approach for SRAM arrays, while [17] proposed varying-sigma sampling with multi-fidelity modeling. However, these approaches share several limitations: they often rely on estimation variance without considering its relative impact on yield, implement coarse precision control

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with limited discrete levels, inadequately capture relationships between precision levels, and frequently target specific circuits or estimators, limiting their scalability.

To address these challenges, we propose Precision-Aware Yield Optimization (PAYO), an efficient framework that conducts yield optimization based on low-precision estimation to reduce computational cost. The novelty of this work includes:

- We develop the first yield optimization framework with precision-aware resource allocation. As shown in Fig. 1, PAYO enables significant reduction in simulation costs through intelligent budget distribution.
- To achieve precise resource allocation, we design a novel precision-based autoregressive model that effectively captures the convergence behavior between true yield and estimation precision.
- The proposed framework is general-purpose and integrates seamlessly with existing sampling-based yield estimation methods through FoM-based precision control.
- We propose an advanced optimization strategy that encodes the quantitative relationship between FoM and simulation cost, balancing information loss and achieving optimal resource allocation for each candidate design.
- Experiments on four benchmark circuits, including a 12nm FinFET In-memory-Computing Vector Multiplier, demonstrate that PAYO reduces simulation costs by more than 10 $\times$ while achieving better optimization results compared to other state-of-the-art (SOTA) methods.

II. BACKGROUND

A. Problem Formulation: Yield Optimization

Let $\mathbf{x} = [x_1, x_2, \dots, x_{d_x}]^T \in \mathcal{X}$ denotes design parameters which are controlled by designer, where $\mathcal{X}$ defines the feasible design space. And the inevitable manufacturing process variations are assumed fully captured by $\mathbf{v} = [v_1, v_2, \dots, v_{d_v}]^T \in \mathcal{V}$. Through normalization, $\mathbf{v}$ can be modeled as a standard independent multivariate Gaussian distribution, where $p(v_i) = \exp(-v_i^2/2) / \sqrt{2\pi}$. Circuit performance metrics m_k depend on both design and variational parameters through the relationship $m_k = h_k(\mathbf{x}, \mathbf{v})$. A circuit is considered qualified when all K performance metrics meet their specified criteria $\mathbf{m}^0$, e.g., $m_k \leq m_k^0$ for $k = 1, 2, \dots, K$. We define an identity function $I : \mathcal{X} \times \mathcal{V} \rightarrow \{0, 1\}$ to indicate whether a design meets these criteria. For a given design $\mathbf{x}$, the yield represents the probability that a manufactured circuit will meet all specifications, expressed mathematically as:

$$g(\mathbf{x}) \triangleq \mathbb{E}(I(\mathbf{x}, \mathbf{v})) = \int_{\mathcal{V}} I(\mathbf{x}, \mathbf{v}) p(\mathbf{v}) d\mathbf{v} \quad (1)$$

The yield optimization problem aims to find the optimal design parameters $\mathbf{x}^*$ that maximize yield:

$$\mathbf{x}^* = \operatorname{argmax}_{\mathbf{x} \in \mathcal{X}} g(\mathbf{x}) \quad (2)$$

This optimization presents two fundamental challenges: the yield function Eq. (1) requires extensive simulation for estimation, and the lack of gradient information $\nabla_{\mathbf{x}} g(\mathbf{x})$ prevents the use of efficient gradient-based optimization methods.

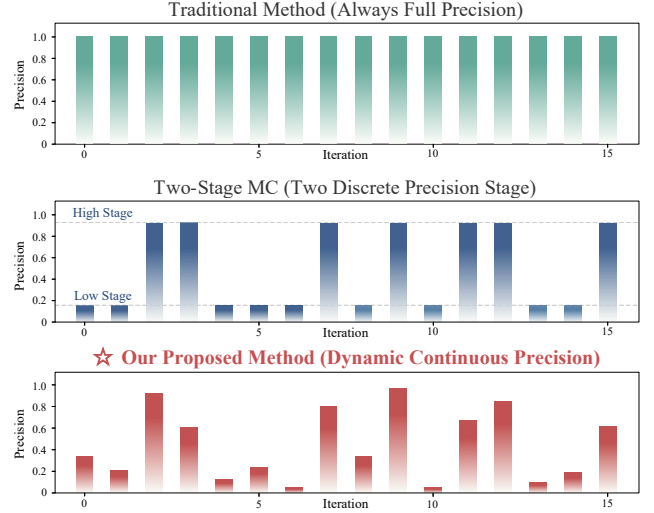


Fig. 1. Illustration of precision allocation among traditional method, Two-stage MC, and our proposed method.

B. Bayesian Yield Optimization

Bayesian optimization provides an efficient framework for optimizing $g(\mathbf{x})$ under these constraints. Since yield estimation relies on random sampling, each estimate contains noise, giving observations of the form $y_i = g(\mathbf{x}_i) + \epsilon_i$, where ϵ_i represents estimation noise. To enable optimization without gradient information, we model $g(\mathbf{x})$ using a GP surrogate model that updates throughout the optimization process. The GP places a prior distribution $g(\mathbf{x}) | \boldsymbol{\theta} \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}') | \boldsymbol{\theta})$ with mean function $m(\mathbf{x}) = \mathbb{E}[g(\mathbf{x})]$ and covariance function $k(\mathbf{x}, \mathbf{x}') = \mathbb{E}[(g(\mathbf{x}) - m(\mathbf{x}))(g(\mathbf{x}') - m(\mathbf{x}'))]$. The hyperparameters $\boldsymbol{\theta}$ define the kernel function and are estimated at each iteration using maximum likelihood estimation. The GP model provides predictive mean, $\mu(\mathbf{x}) = \mathbf{k}(\mathbf{x})^T (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}$ and variance $v(\mathbf{x}) = \sigma^2 + k(\mathbf{x}, \mathbf{x}) - \mathbf{k}^T(\mathbf{x}) (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}(\mathbf{x})$. Given the high computational cost of yield estimation, selecting the next design point $\mathbf{x}_t$ for evaluation requires careful balance between exploiting regions of high observed yield and exploring uncertain regions. This balance is achieved through an acquisition function, with Upper Confidence Bound (UCB) being particularly effective:

$$\mathbf{x}_t = \operatorname{argmax}_{\mathbf{x} \in \mathcal{X}} \left(\mu(\mathbf{x}) + \beta^{\frac{1}{2}} v(\mathbf{x}) \right) \quad (3)$$

where β controls the exploration-exploitation trade-off.

C. Multi-Fidelity Bayesian Optimization

Multi-fidelity Bayesian optimization (MFBO) extends traditional BO by leveraging evaluations of varying computational cost and accuracy. In yield optimization, higher fidelity evaluations require more simulations, providing more accurate yield estimates but at greater computational cost. Let function $f(\mathbf{x}, z)$ represent the estimated yield for design $\mathbf{x}$ at fidelity $z \in [z^0, z^1]$, where z^0 denotes the lowest fidelity (fewer simulations) and z^1 the highest fidelity (maximum simulations). The MFBO framework can be formally described as follows:

$$\mathbf{x}^* = \operatorname{argmax}_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}, z^1) \text{ s.t. } \sum_{t=1}^T c(z_t) \leq B \quad (4)$$

where $c(z_t)$ represents the computational cost at fidelity level z_t (assuming independent of $\mathbf{x}$), T is the total number of

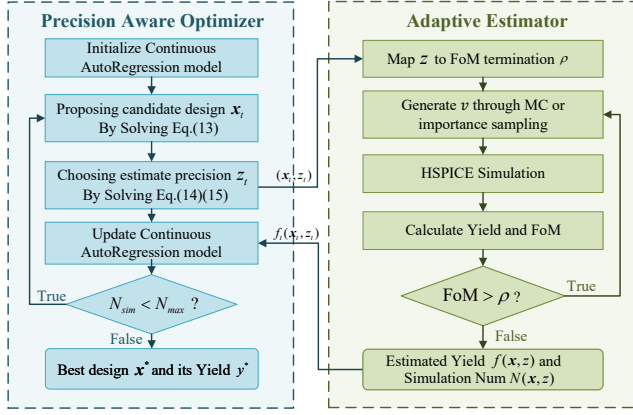


Fig. 2. The workflow of our proposed Precision-Aware Yield Optimization.

evaluations, and B is the total computational budget. Generally speaking, at each iteration, MFBO selects both the design point and fidelity level by maximizing a multi-fidelity acquisition function:

$$(\mathbf{x}_t, z_t) = \underset{\mathbf{x} \in \mathcal{X}, z \in \mathcal{Z}}{\operatorname{argmax}} \alpha(\mathbf{x}, z) / c(z) \quad (5)$$

where $\alpha(\mathbf{x}, z)$ is the acquisition function value and $c(z)$ represents the evaluation cost at fidelity z . Most existing MFBO frameworks focus on discrete fidelity levels due to their simpler implementation. However, this discretization can limit the efficiency of resource allocation, particularly in yield optimization where the relationship between simulation cost and estimation accuracy is inherently continuous. This limitation motivates our development of a continuous fidelity approach specifically designed for yield optimization.

III. PROPOSED APPROACH

To address the limitations of existing yield optimization approaches, we propose PAYO (Precision-Aware Yield Optimization), a framework that intelligently allocates computational resources by dynamically adjusting yield estimation precision. Our framework consists of three key components: (1) an adaptive yield estimator that provides continuous precision control through Figure of Merit (FoM), (2) a continuous autoregressive model that captures the relationship between different precision levels, and (3) a precision-aware optimization strategy that efficiently balances exploration and exploitation across the precision spectrum.

A. Framework Overview

Fig. 2 illustrates the PAYO workflow: In a single iteration, the precision-aware optimizer first selects a promising design candidate $\mathbf{x}_t$ and its ideal precision level z_t with potential maximum gain. Adaptive yield estimation is then conducted accordingly. All data are collected to train a continuous autoregressive model to build a relation between truth yield and low-precision yield analysis. With these three components integrated, yield analyses are conducted at a “just fine” precision to reduce computational cost in yield optimization.

B. Adaptive Yield Estimator with Continuous Precision

In precision-aware yield optimization, a critical challenge is mapping the desired precision level z_t to an appropriate simulation count N . Classic approaches often employ simple

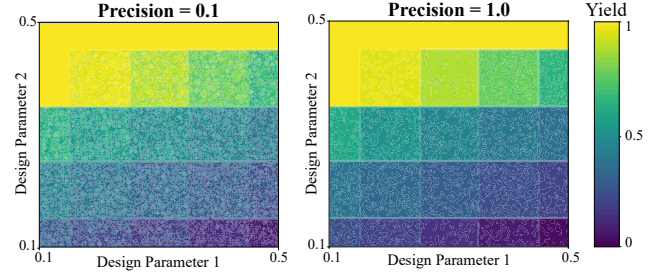


Fig. 3. Yield distribution for OTA circuit: Low precision estimation does not significantly influence the yield trend and is sufficient to guide optimization.

strategies which proved to be inadequate in practice. The most straightforward approach uses linear mapping: $N = N_{\max} z_t$, where $N_{\max}$ represents the maximum simulation count per iteration. However, this naive approach fails to account for the varying difficulty in estimating yields of different magnitudes. For instance, accurately estimating an extremely high yield (i.e., very low failure rate P_f) requires substantially more simulations than estimating a moderate yield value. Several works [12], [13], [15] propose variance-based resource allocation using the approximation:

$$N \approx \frac{Y \cdot (1 - Y) \cdot k_\gamma^2}{\Delta Y^2} \quad (6)$$

where ΔY defines the confidence interval and k_γ reflects the confidence level. However, this approach suffers from two fundamental limitations: (1) Its derivation assumes the binomial distribution can be approximated by a normal distribution when N is large enough—an invalid assumption for the typically small sample sizes used in yield optimization. (2) It treats estimation variance uniformly across different yield levels, ignoring that the same variance has drastically different implications for different failure rates (e.g., a variance of 0.1 is far more significant for $P_f = 10^{-8}$ than for $P_f = 0.5$).

To address these limitations, we adopt FoM as a more effective precision measure. For Monte Carlo methods, FoM quantifies the relative error in yield estimation:

$$\rho = \frac{\sqrt{\operatorname{Var}(\hat{P}_f)}}{\hat{P}_f} = \sqrt{\frac{1}{N} \cdot \frac{1 - \hat{P}_f}{\hat{P}_f}} \quad (7)$$

For importance sampling methods, FoM is defined as:

$$\rho = \frac{\sqrt{\operatorname{Var}(\hat{P}_f)}}{\hat{P}_f} = \frac{1}{N} \cdot \sqrt{\frac{\sum_{i=1}^N w(\mathbf{v}_i)^2 I(\mathbf{x}, \mathbf{v})}{\hat{P}_f^2} - N} \quad (8)$$

where $\hat{P}_f$ represents the estimated failure rate and $w(\mathbf{v}_i)$ denotes importance sampling weights. Building on this FoM foundation, we propose an adaptive yield estimator that dynamically controls precision through continuous FoM monitoring. During estimation, we compute the current FoM after each sampling round and continue sampling until FoM drops below $1 - 0.9z_t$. This approach provides several key advantages:

Natural Resource Allocation: As evident from Eq. (7) and Eq. (8), designs with lower failure rates automatically receive more simulation resources, aligning with the need for higher precision during exploitation phases.

Efficient Low-Precision Evaluation: The quadratic relationship between FoM and simulation count means that low-

precision evaluations require minimal resources. As demonstrated in Fig. 3, these evaluations can still effectively capture yield trends, making them ideal for exploration phases.

General Applicability: The estimator seamlessly integrates with both MC and importance sampling methods through their respective FoM definitions, providing a unified framework for precision control.

This adaptive approach ensures that computational resources are allocated based on both the desired precision level and the inherent difficulty of yield estimation for each design point, enabling more efficient optimization compared to traditional fixed-allocation strategies.

C. Continuous AutoRegression for Precision Modeling

Traditional multi-fidelity yield optimization approaches [16], [17] typically treat fidelity as an independent input dimension in their GP models. While straightforward to implement, this approach fails to capture the inherent convergence behavior of yield estimation—as precision increases, estimates gradually converge to their true values. Furthermore, treating precision levels independently ignores the strong correlations between successive precision levels, leading to inefficient information utilization during optimization.

To address these limitations, we employ a Continuous AutoRegression model [18] with a specially designed kernel structure that captures the precision-dependent nature of yield estimation. Our key insight is that yield estimates at different precision levels exhibit a continuous evolution pattern that can be modeled through differential equations. We formulate this relationship using a differential equation:

$$\frac{\partial f(\mathbf{x}, z)}{\partial z} = \alpha(z)f(\mathbf{x}, z) + s(\mathbf{x}, z) \quad (9)$$

where $f(\mathbf{x}, z)$ represents the yield estimate at precision z , $\alpha(z)$ controls the convergence rate, and $s(\mathbf{x}, z)$ captures additional nonlinear relationships. By setting $\alpha(z)$ to a positive constant, we ensure the model reflects the natural convergence behavior of yield estimation. The solution to this differential equation is:

$$f(\mathbf{x}, z) = f(\mathbf{x}, z^0)e^{-\alpha(z-z^0)} + \int_{z^0}^z e^{-\alpha(z-\tau)} s(\mathbf{x}, \tau) d\tau \quad (10)$$

where z^0 represents the lowest precision level. We place GP priors on both the base function and the residual term: $f(\mathbf{x}, z^0) \sim \mathcal{GP}(0, k_0(x, x'))$ and $s(\mathbf{x}, z) \sim \mathcal{GP}(0, k_s((x, z), (x', z')))$. This formulation leads to a precision-aware kernel structure:

$$k((\mathbf{x}, z), (\mathbf{x}', z')) = e^{-\alpha(z-z^0)} k_0(\mathbf{x}, \mathbf{x}') + \int_{z^0}^z e^{-\alpha(z-\tau)} \int_{z^0}^{\tau} e^{-\alpha(\tau'-\tau')} k_s((\mathbf{x}, \tau), (\mathbf{x}', \tau')) d\tau' d\tau, \quad (11)$$

which explicitly models the convergence behavior through the exponential terms, captures both design-space relationships (k_0) and precision-level correlations (k_s), and enables efficient information transfer across precision levels through the continuous formulation.

The combination of continuous autoregression and precision-aware kernel structure allows our model to better capture the yield estimation behavior across different

precision levels, leading to more efficient resource allocation and optimization convergence.

D. Precision-Aware Bayesian Yield Optimization

With the autoregressive model capturing the relationship between different precision levels, we can now leverage this information for efficient yield optimization. Unlike traditional GP models that treat fidelity as an independent dimension, our autoregressive model provides rich information about the convergence behavior of yield estimates, enabling more informed decisions about precision selection. Multi-fidelity Bayesian optimization typically selects both design parameters and fidelity levels by maximizing an acquisition function augmented with fidelity:

$$\mathbf{x}_t, z_t = \operatorname{argmax}_{\mathbf{x} \in \mathcal{X}, z \in \mathcal{Z}} \left(\mu(\mathbf{x}, z) + \beta(z)^{\frac{1}{2}} v(\mathbf{x}, z) \right) \quad (12)$$

However, this approach has limitations for yield optimization: it doesn't account for the unique characteristics of yield estimation precision, and it may waste resources by evaluating designs at unnecessarily high precision levels. To address these limitations, we propose a novel optimization strategy that leverages our autoregressive model and FoM-based precision control inspired by [19]. First, we select promising designs based on their potential at full precision:

$$\mathbf{x}_t = \operatorname{argmax}_{\mathbf{x} \in \mathcal{X}} \left(\mu(\mathbf{x}, z^1) + \beta_t^{\frac{1}{2}} v(\mathbf{x}, z^1) \right) \quad (13)$$

where β_t increases with iterations to maintain exploration. Then, we determine the appropriate precision level through a novel selection strategy:

$$z_t = \operatorname{argmin}_{z \in \mathcal{Z}_t(\mathbf{x}_t) \cup z^1} N(\mathbf{x}_t, z) \quad (14)$$

subject to precision constraints:

$$\mathcal{Z}_t(\mathbf{x}_t) = \{z \in \mathcal{Z} \setminus \{z^1\} : v(\mathbf{x}_t, z) > \sqrt{\kappa_0} \xi(z) \left(\frac{1 - 0.9z^1}{1 - 0.9z} \right)^{2q}, \xi(z) > \beta_t^{-1/2} \|\xi\|_{\infty}\} \quad (15)$$

where κ_0 is the length scale of kernel function. q is a constant dependent on kernel used for z , e.g., for the squared exponential kernel, $q = 1/(d_x + 3)$.

This strategy incorporates several key innovations:

Information-Cost Balance: The first constraint balances information gain (measured by $\xi(z) = \sqrt{1 - \phi_z(z, z^*)^2}$ where ϕ_z is the kernel function of z) against computational cost, leveraging the Continuous AutoRegression model's understanding of precision relationships. Furthermore, the quadratic relationship between FoM and simulation count is taken into account to make more reasonable precision selection.

Efficiency Threshold: The second constraint prevents the selection of unnecessarily high precision levels by setting an exclusion zone near full precision.

Adaptive Resource Allocation: The framework automatically adjusts precision based on: (1) exploration needs (lower precision for initial search) (2) exploitation requirements (higher precision for promising regions) (3) convergence behavior captured by the Continuous AutoRegression model.

As shown in Algorithm 1, the complete PAYO algorithm integrates these components, providing a comprehensive framework for precision-aware yield optimization.

TABLE I
YIELD OPTIMIZATION FOR DIGITAL CIRCUITS WITH DIFFERENT NUMBERS OF TOTAL SIMULATIONS

Circuit	Method	# Simulation=1,000				# Simulation=5,000				# Simulation=10,000			
		Worst	Best	Mean	Std	Worst	Best	Mean	Std	Worst	Best	Mean	Std
SRAM (Fail rate)	AYO	4.14e-1	4.13e-1	4.13e-1	8.28e-5	2.50e-7	5.00e-8	1.00e-7	8.66e-8	1.50e-7	5.00e-8	8.00e-8	4.47e-8
	KDEBO	4.11e-1	3.77e-2	2.83e-1	1.49e-1	3.93e-1	4.73e-6	1.25e-1	1.58e-1	4.95e-6	1.92e-8	2.12e-6	2.49e-6
	MVEBO	4.14e-1	2.73e-3	3.31e-1	1.84e-1	4.13e-1	2.94e-8	9.31e-2	1.80e-1	2.50e-7	1.03e-8	7.04e-8	1.02e-7
	BYABO	4.13e-1	4.13e-1	4.13e-1	1.52e-4	1.85e-6	1.32e-8	1.30e-6	7.50e-7	1.70e-6	1.32e-8	5.43e-7	7.47e-7
	OPT	1.08e-1	1.08e-1	1.08e-1	6.22e-5	9.30e-8	1.05e-8	4.73e-8	3.49e-8	9.30e-8	1.05e-8	4.73e-8	3.49e-8
	PAYO-IS	9.30e-8	1.61e-8	5.12e-8	3.26e-8	9.30e-8	1.61e-8	5.12e-8	3.26e-8	9.30e-8	1.61e-8	5.12e-8	3.26e-8
	PAYO-MC	9.30e-8	1.61e-8	4.52e-8	2.94e-8	9.30e-8	1.61e-8	4.52e-8	2.94e-8	9.30e-8	1.61e-8	4.52e-8	2.94e-8
ADDER (Fail rate)	AYO	5.68e-3	5.63e-3	5.66e-3	1.65e-5	1.05e-7	2.94e-8	7.39e-8	2.99e-8	1.05e-7	1.92e-8	4.62e-8	3.46e-8
	KDEBO	8.20e-2	4.76e-8	4.94e-2	4.45e-2	5.88e-8	2.56e-8	4.52e-8	1.44e-8	5.88e-8	2.50e-8	3.85e-8	1.44e-8
	MVEBO	5.68e-3	5.63e-3	5.66e-3	1.69e-5	9.30e-8	1.56e-8	5.78e-8	3.12e-8	9.30e-8	1.56e-8	3.68e-8	3.19e-8
	BYABO	5.68e-3	5.64e-3	5.66e-3	1.76e-5	8.70e-8	2.94e-8	4.46e-8	2.50e-8	2.94e-8	1.75e-8	2.47e-8	6.42e-9
	OPT	7.15e-2	7.13e-2	7.14e-2	6.06e-5	3.49e-8	1.82e-8	2.84e-8	6.13e-9	3.49e-8	1.82e-8	2.84e-8	6.13e-9
	PAYO-IS	7.14e-8	1.82e-8	4.02e-8	1.92e-8	7.14e-8	1.82e-8	4.02e-8	1.92e-8	7.14e-8	1.82e-8	4.02e-8	1.92e-8
	PAYO-MC	2.94e-8	8.70e-9	2.30e-8	8.37e-9	2.94e-8	8.70e-9	2.30e-8	8.37e-9	2.94e-8	8.70e-9	2.30e-8	8.37e-9

Algorithm 1 Precision-Aware Bayesian Yield Optimization

Input: Adaptive yield estimator $f(\mathbf{x}, z)$, # Initial samples N_i , # BO iteration N_o

- 1: Sample $\{\mathbf{x}_i, z_i\}, i = 1, \dots, N_i$ to estimate the yield y_i
- 2: Set the initial observation set $\mathcal{D}_i \leftarrow \{y_i, \mathbf{x}_i, z_i\}, i = 1, \dots, N_i$
- 3: Initialize Continuous AutoRegression model with initial $\mathcal{D}$
- 4: **for** $t = 1$ to N_o **do**
- 5: Propose candidate design $\mathbf{x}_t$ by solving Eq. (13)
- 6: Select precision z_t by solving Eq. (14) and Eq. (15)
- 7: Set initial FoM $\rho = \infty$
- 8: **while** $\rho > 1 - 0.9z_t$ **do**
- 9: Randomly sample variation parameters $\mathbf{v}$
- 10: Obtain performance metric by SPICE simulation
- 11: Update the yield y_t and FoM ρ
- 12: **end while**
- 13: $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{y_t, \mathbf{x}_t, z_t\}$
- 14: Update the Continuous AutoRegression model
- 15: **end for**
- 16: **return** Best design parameter $\mathbf{x}^*$

IV. EXPERIMENTAL RESULTS

To evaluate the efficiency and optimization performance of our proposed framework, we conducted experiments on four benchmark circuits: a 6T-SRAM, an Adder, an Operational Transconductance Amplifier (OTA), and an In-Memory-Computing Vector Multiplier (IMC-VM). These benchmark circuits encompass both digital and analog designs, as well as advanced and traditional process nodes, allowing for a comprehensive assessment of PAYO's scalability and robustness across different circuit paradigms and technology nodes. To ensure rigorous evaluation, we implement five SOTA methods as comparative baselines: AYO [12], KDEBO [14], MVEBO [13], BYABO [6], and OPT [8]. To eliminate the influence of randomness, each method was tested using five random seeds and the optimal design was validated using $10e7$ MC simulations (IMC-VM using $10e3$). All experiments are conducted on a Linux system equipped with an Intel Xeon Platinum 8352V CPU, and 60GB of memory.

A. Yield Optimization on Digital Circuit

The performance of our proposed framework in digital circuit yield optimization was validated using two circuits: 6T-SRAM and Adder, both of which are implemented using 45nm CMOS process. Fig. 4 (a) shows the schematic of the 6T-SRAM, which consists of a latch that can stably store 1bit data and two access transistors for read/write operation. We consider the read/write delay as the performance metric and the widths of the three symmetric transistor couples are chosen as the design parameters of interest. As depicted in Fig. 4 (b), Adder circuit consists of 28 transistors, the widths and lengths of transistors are chosen as the design parameters. We compute the yield by the delay of the output reaching a predefined threshold voltage through transient simulations.

As shown in Table. I, it is evident that PAYO achieved the best optimization results with the minimal simulation cost. For 6T-SRAM, PAYO-MC achieves a performance improvement of $2.39e6 \times - 9.14e6 \times$ with 1,000 simulations, $1.05 \times - 2.77e6 \times$ improvement with 5,000 simulations, and $1.05 \times - 46.90 \times$ improvement with 10,000 simulations. For Adder, PAYO-MC achieves $2.46e5 \times - 3.10e6 \times$ improvement with 1,000 simulations, $1.23 \times - 3.21 \times$ improvement with 5,000 simulations, and $1.07 \times - 2.01 \times$ improvement with 10,000 simulations. To verify the compatibility of PAYO with importance sampling methods, we embedded Minimal Norm Importance Sampling (MNIS) [1] into the PAYO framework for experimentation (PAYO-IS). Due to the lower accuracy and reliability of MNIS compared to MC, PAYO-IS performs slightly worse than PAYO-MC. However, it still achieves better performance than other baselines.

B. Yield Optimization on Analog circuit

To validate the effectiveness of our proposed method for analog circuits, we conducted experiment on the OTA and IMC-VM circuit. The OTA, implemented in 180nm CMOS proces, consists of a bias circuit and a two-stage amplifier with miller compensation, as shown in Fig. 4 (c). The OTA's static current is considered as the target performance metric

TABLE II
YIELD OPTIMIZATION FOR ANALOG CIRCUITS WITH DIFFERENT NUMBERS OF TOTAL SIMULATIONS

Circuit	Method	# Simulation=1,000				# Simulation=5,000				# Simulation=10,000			
		Worst	Best	Mean	Std	Worst	Best	Mean	Std	Worst	Best	Mean	Std
OTA (Fail rate)	AYO	5.29e-1	5.29e-1	5.29e-1	1.00e-4	1.46e-2	8.82e-8	8.77e-3	8.00e-3	8.82e-8	1.00e-8	3.56e-8	3.05e-8
	KDEBO	5.82e-1	1.46e-2	3.18e-1	2.18e-1	3.99e-1	1.33e-5	1.22e-1	1.76e-1	5.88e-8	1.75e-8	3.47e-8	1.84e-8
	MVEBO	1.46e-2	7.14e-8	4.02e-3	6.08e-3	6.52e-8	1.35e-8	3.70e-8	1.99e-8	6.52e-8	1.25e-8	3.13e-8	2.15e-8
	BYABO	5.29e-1	5.29e-1	5.29e-1	4.55e-5	1.40e-5	2.00e-8	7.96e-6	7.24e-6	8.82e-8	1.89e-8	3.51e-8	2.99e-8
	PAYO-MC	4.08e-8	1.15e-8	2.55e-8	1.18e-8	4.08e-8	1.15e-8	2.55e-8	1.18e-8	4.08e-8	1.15e-8	2.55e-8	1.18e-8
IMC-VM (Yield)	AYO	93.9%	94.3%	94.2%	0.15%	95.4%	99.7%	98.0%	1.57%	99.1%	99.7%	99.3%	0.20%
	KDEBO	77.4%	99.2%	91.9%	7.69%	94.2%	99.6%	97.4%	1.99%	97.6%	99.7%	99.0%	0.76%
	MVEBO	94.2%	94.4%	94.3%	0.08%	94.3%	99.6%	98.0%	1.92%	98.7%	99.7%	99.3%	0.35%
	BYABO	82.1%	99.4%	92.1%	6.73%	96.7%	99.4%	98.5%	0.96%	98.1%	99.5%	99.0%	0.53%
	PAYO-MC	97.6%	99.7%	98.8%	0.72%	98.9%	99.7%	99.4%	0.28%	98.9%	99.7%	99.4%	0.28%

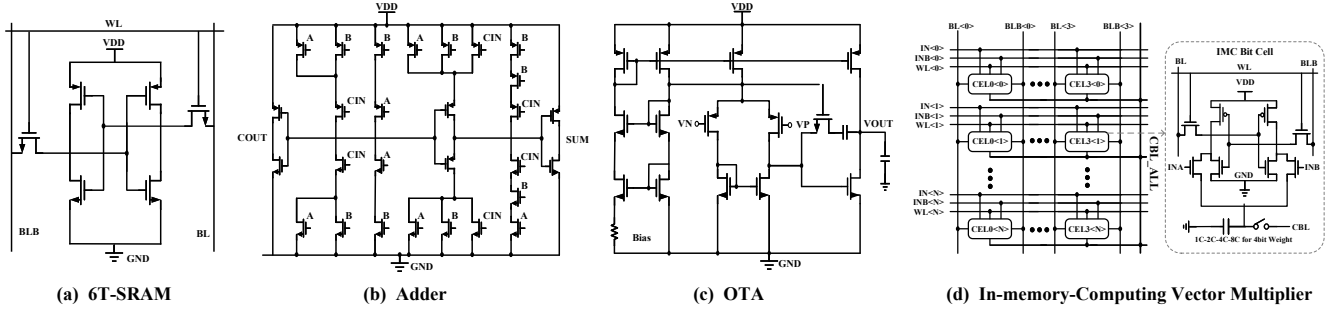


Fig. 4. Simplified schematic for benchmark circuits: 6T-SRAM, Adder, OTA and IMC Vector Multiplier

and the widths of current mirrors and bias transistors are selected as the design parameters for yield optimization. Fig. 4 (d) shows the structure of the IMC-VM proposed by [20], where each input is multiplied with the 4bit weight stored in the 9T-1C IMC cell and accumulated on the global charge bit line through charge sharing. Multibit-weight modulation is conducted through capacitance difference. Sum of each node's settling time of is selected as the performance metric. As the circuit is implemented in 12nm FinFET process, the number of fins and fingers of each FinFET in the 9T-1C cell is considered as the design parameter of interest.

Due to the poor robustness of OPT and MNIS in complex analog circuit yield analysis, We did not include OPT and PAYO-IS in the comparison. As shown in Table. II, our method also demonstrates the best performance among all baselines on the analog circuits. For the OTA, PAYO-MC achieved performance improvements of $1.58e5 \times$ – $2.07e7 \times$ with 1000 simulations, $1.45 \times$ – $4.78e6 \times$ with 5000 simulations, and $1.23 \times$ – $1.40 \times$ with 10000 simulations. For the IMC Vector Multiplier, PAYO-MC achieved performance improvements of 4.77%–7.51%, 0.91%–2.05%, and 0.10%–0.40% with 1000, 5000, and 10000 simulations, respectively.

C. Ablation Study

To validate the effectiveness of each component in PAYO, we conducted ablation experiments on the Adder circuit. As shown in Fig. 5, we compared the number of simulations required by each method to first find a design with a failure rate less than $1e-7$. Each method was set with $N_{\max} = 2000$. Compared to full precision, linear mapping $N = N_{\max} z_t$, and Two-Stage MC, our proposed FoM mapping method achieved

improvement factors of $11.28 \times$, $9.99 \times$, and $4.54 \times$, respectively. Furthermore, compared to GP regression (GPR), the Continuous AutoRegression (CAR) model we used improved optimization efficiency by $1.36 \times$.

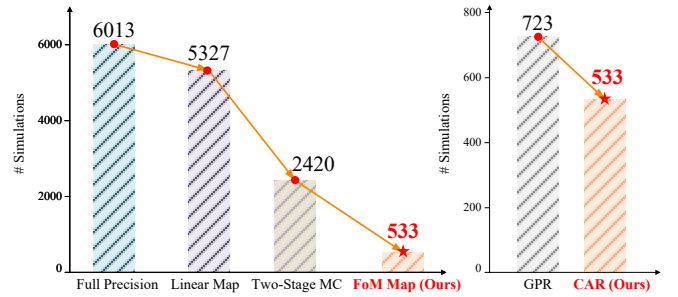


Fig. 5. Ablation study of each component.

V. CONCLUSION

This paper presents PAYO, the first precision-aware yield optimization framework that significantly reduces simulation costs (more than $10 \times$ reduction). Key innovations include FoM-based precision control, continuous autoregressive modeling for precision relationships, and an efficient resource allocation strategy. Current limitations include computational overhead from the autoregression model and the need for initial samples across different precision levels. Future work will explore extensions to multi-objective yield optimization and online learning strategies for more efficient precision control.

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